

Lecture 11 -- Gradient Descent

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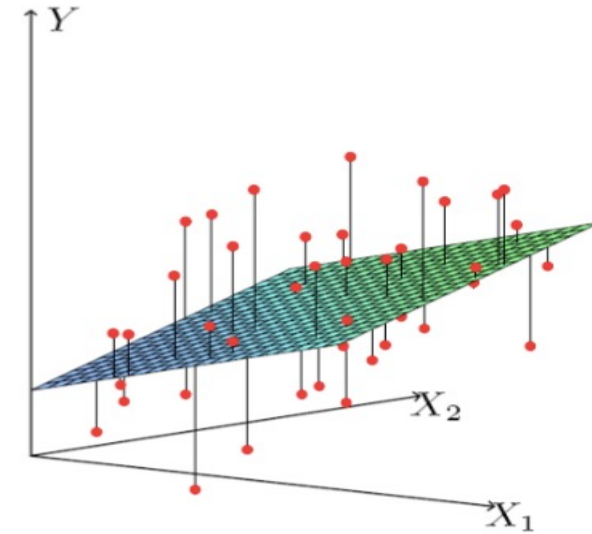
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Up to now,

- Three machine learning algorithms:
 - decision tree learning
 - k-nn
 - linear regression
 - linear regression
 - linear classification
 - logistic regression

Discussed how to formulate objective functions but didn't explain how to solve/optimize.



Today's Topics

Problem Statement

foundations

Derivative and Gradient

Main method

Method of Gradient Descent

Example: Gradient Descent on Linear Regression

Linear Regression: Analytical Solution

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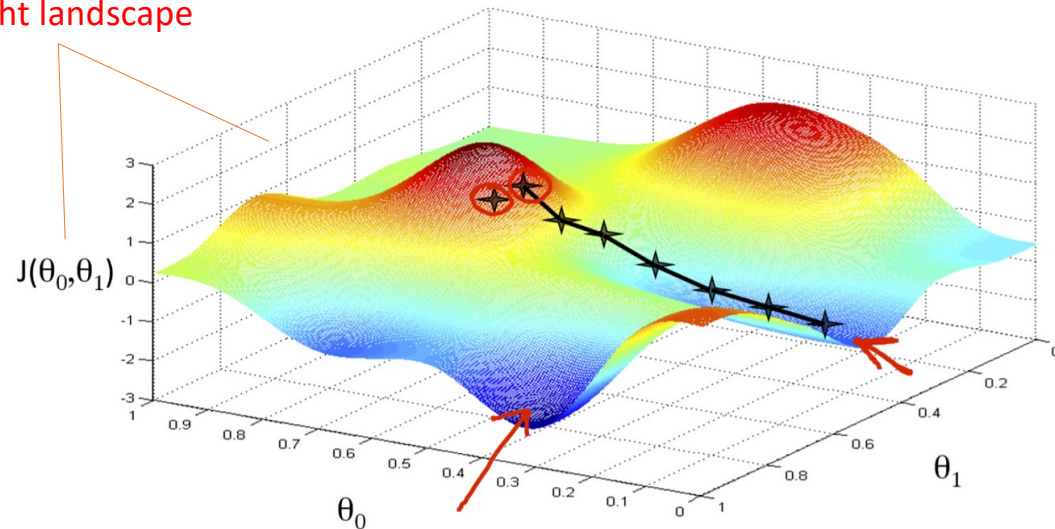
An example of how to design optimisation algorithm

Problem Statement: Gradient-Based Optimization

Why challenging?

1. We **do not see** the landscape;
2. We need to make algorithms to **automatically and smartly** search for the optimum

Weight landscape

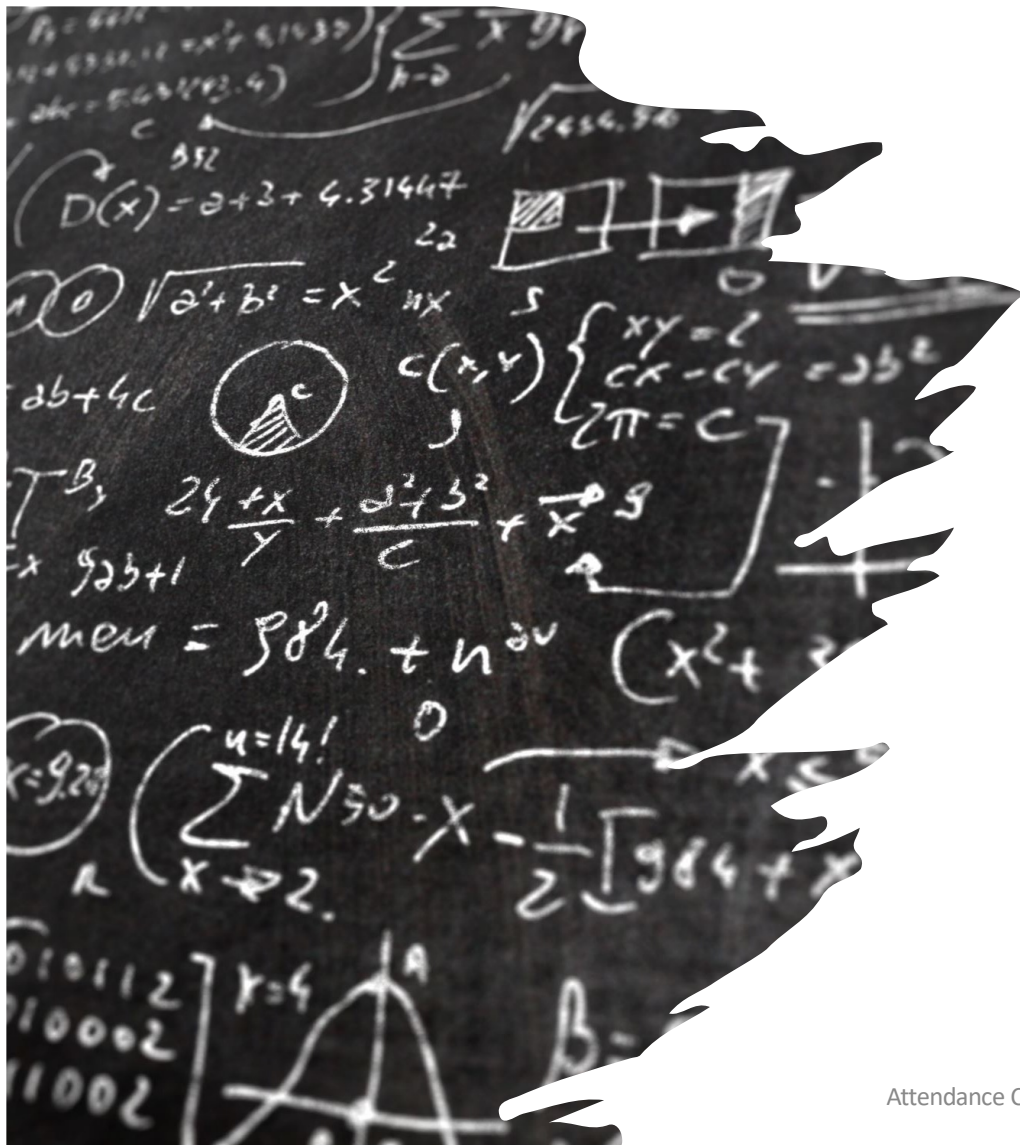


Problem Statement: Gradient-Based Optimization

- Most ML algorithms involve optimization
- Minimize/maximize a function $f(\mathbf{x})$ by altering $\mathbf{x}$
 - Maximization accomplished by minimizing $-f(\mathbf{x})$
- $f(\mathbf{x})$ referred to as objective function or criterion
 - In minimization also referred to as loss function cost, or error
 - Example:
 - linear least squares $f(x) = \frac{1}{2} \|Ax - b\|^2$
 - Linear regression $\hat{L}(f_w) = \frac{1}{m} \sum_{i=1}^m (w^T x^{(i)} - y^{(i)})^2$
- Denote optimum value by $\mathbf{x}^* = \operatorname{argmin} f(\mathbf{x})$

How? Along the gradient descent direction.

General description of the problem statement



Derivative and Gradient

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Derivative of a function

- Suppose we have function $y=f(x)$, x, y real numbers

- Derivative of function denoted: $f'(x)$ or as dy/dx

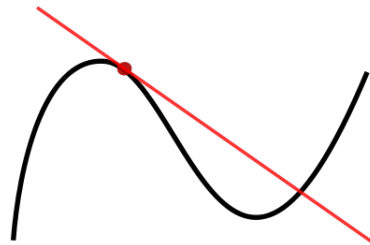
- Derivative $f'(x)$ gives the slope of $f(x)$ at point x

- It specifies how to scale a small change in input to obtain a corresponding change in the output:

$$f(x + \Delta) \approx f(x) + \Delta f'(x)$$

How to design Δ (*learning rate*) ? This is an art of machine learning

- It tells how you make a small change in input to make a small improvement in y



Recall what's the derivative for the following functions:

$$f(x) = x^2$$

$$f(x) = e^x$$

...

Opposite sign of derivative, why?

- Suppose we have function $y = f(x)$, where x, y are real numbers
- Sign function:

$$\text{sign}(x) = \begin{cases} -1 & \text{if } x < 0 \\ 0 & \text{if } x = 0 \\ 1 & \text{if } x > 0 \end{cases}$$

Why does the expression like this?

Example to be given in the next slide

- We know that

$$f(x - \epsilon \text{sign}(f'(x))) < f(x)$$

for small ϵ .

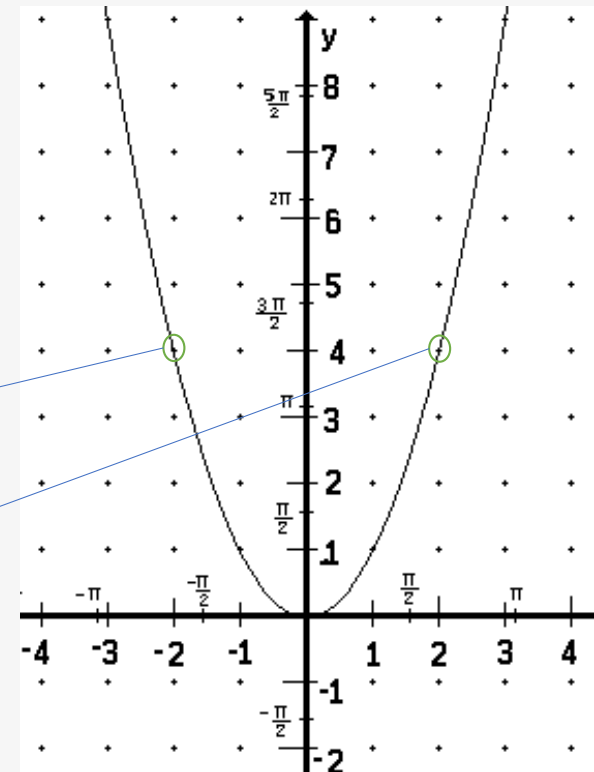
- Therefore, we can reduce $f(x)$ by moving x in small steps with opposite sign of derivative

Example: $\text{sign}((-2,3)) = (-1,1)$

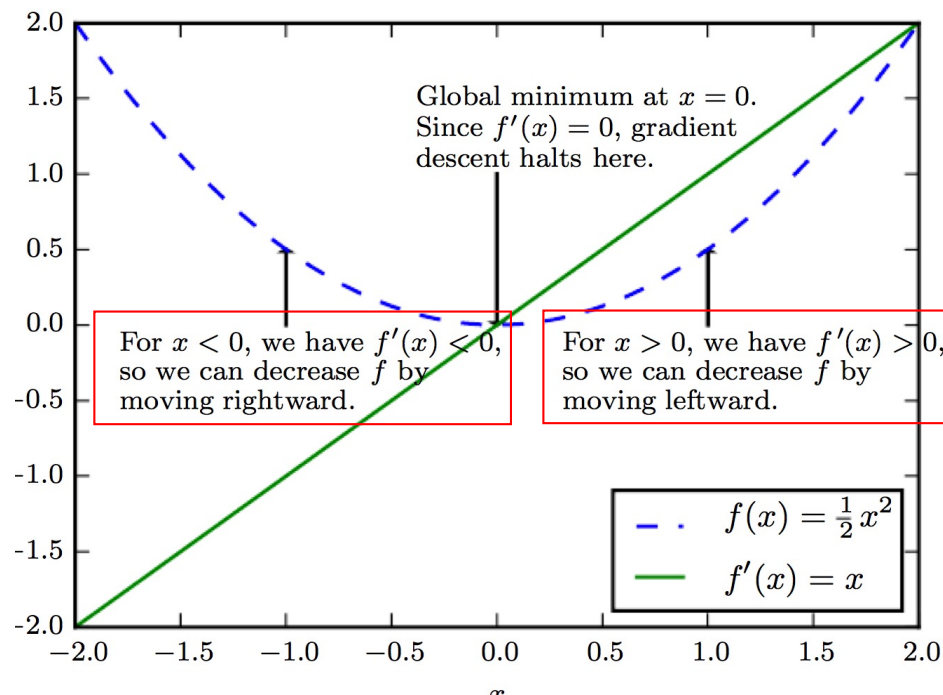
This technique is called **gradient descent** (Cauchy 1847)

Example for $f(x - \epsilon \text{sign}(f'(x))) < f(x)$

- Function $f(x) = x^2$ $\epsilon = 0.1$
- $f'(x) = 2x$
- For $x = -2$, $f'(-2) = -4$, $\text{sign}(f'(-2)) = -1$
- $f(-2 - \epsilon * (-1)) = f(-1.9) < f(-2)$
- For $x = 2$, $f'(2) = 4$, $\text{sign}(f'(2)) = 1$
- $f(2 - \epsilon * 1) = f(1.9) < f(2)$



Opposite sign of derivative, Illustrated



For $x < 0$, $f(x)$ decreases with x and $f'(x) < 0$

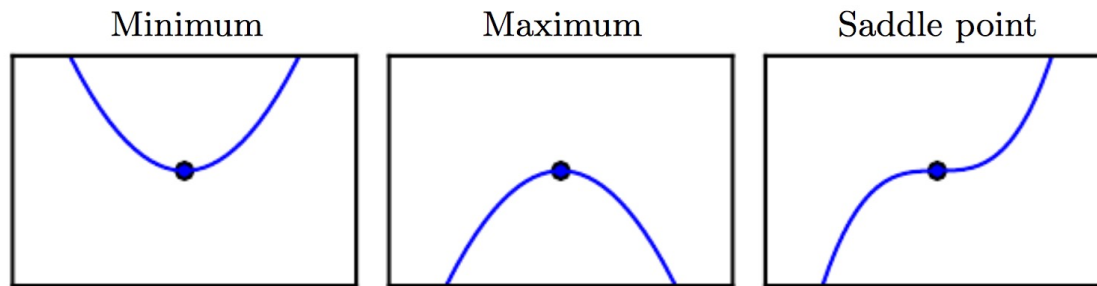
For $x > 0$, $f(x)$ increases with x and $f'(x) > 0$

Use $f'(x)$ to follow function downhill

Reduce $f(x)$ by going in direction opposite sign of derivative $f'(x)$

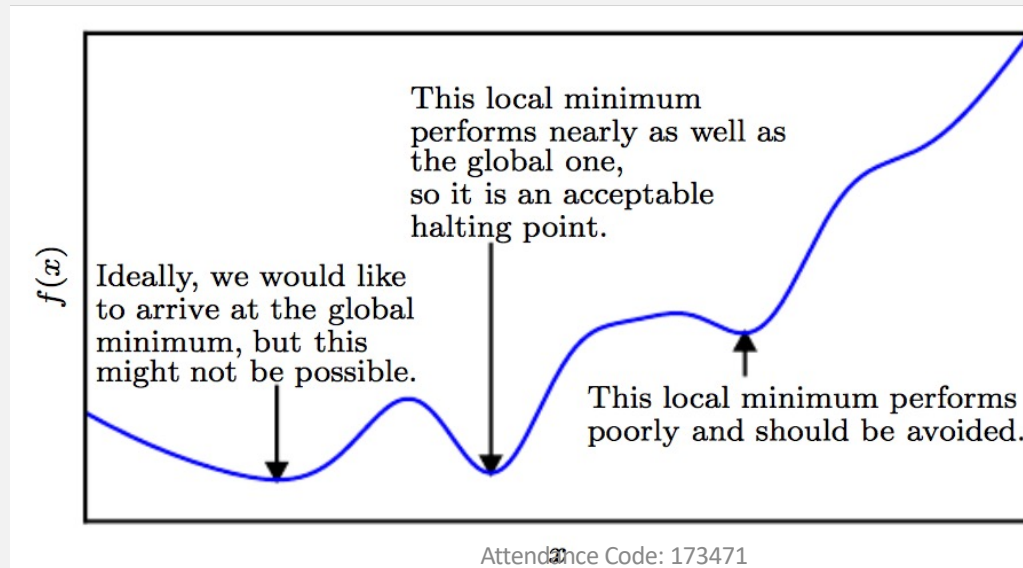
Stationary points, Local Optima

- When $f'(x) = 0$ derivative provides no information about direction of move
- Points where $f'(x) = 0$ are known as *stationary* or critical points
 - Local minimum/maximum: a point where $f(x)$ lower/ higher than all its neighbors
 - Saddle Points: neither maxima nor minima



Presence of multiple minima

- Optimization algorithms may fail to find global minimum
- Generally accept such solutions

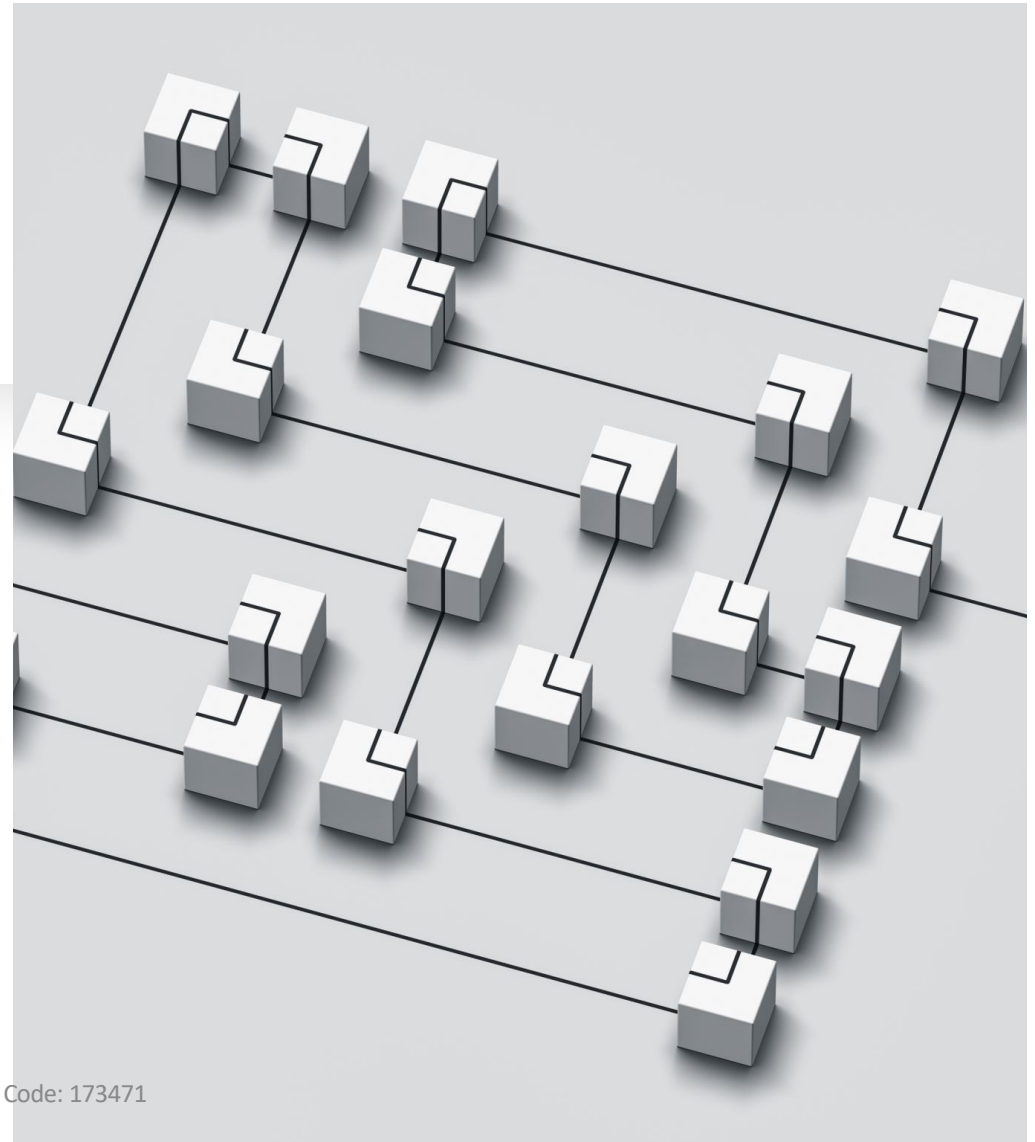


Minimizing with multiple dimensional inputs

- We often minimize functions with multiple-dimensional inputs

$$f : \mathbb{R}^n \rightarrow \mathbb{R}$$

- For minimization to make sense there must still be only one (scalar) output



Functions with multiple inputs

- Partial derivatives

$$\frac{\partial}{\partial x_i} f(x)$$

measures how f changes as only variable x_i increases at point x

- Gradient generalizes notion of derivative where derivative is wrt a vector
- Gradient is vector containing all of the partial derivatives denoted

$$\nabla_x f(x) = \left(\frac{\partial}{\partial x_1} f(x), \dots, \frac{\partial}{\partial x_n} f(x) \right)$$

Example

-
- $y = 5x_1^5 + 4x_2 + x_3^2 + 2$
 - so what is the exact gradient on instance (1,2,3)?
 - the gradient is $(25x_1^4, 4, 2x_3)$
 - On the instance (1,2,3), it is (25,4,6)

Functions with multiple inputs

- **Gradient** is vector containing all of the partial derivatives denoted

$$\nabla_x f(x) = \left(\frac{\partial}{\partial x_1} f(x), \dots, \frac{\partial}{\partial x_n} f(x) \right)$$

- Element i of the gradient is the partial derivative of f wrt x_i
- **Critical points** are where every element of the gradient is equal to zero

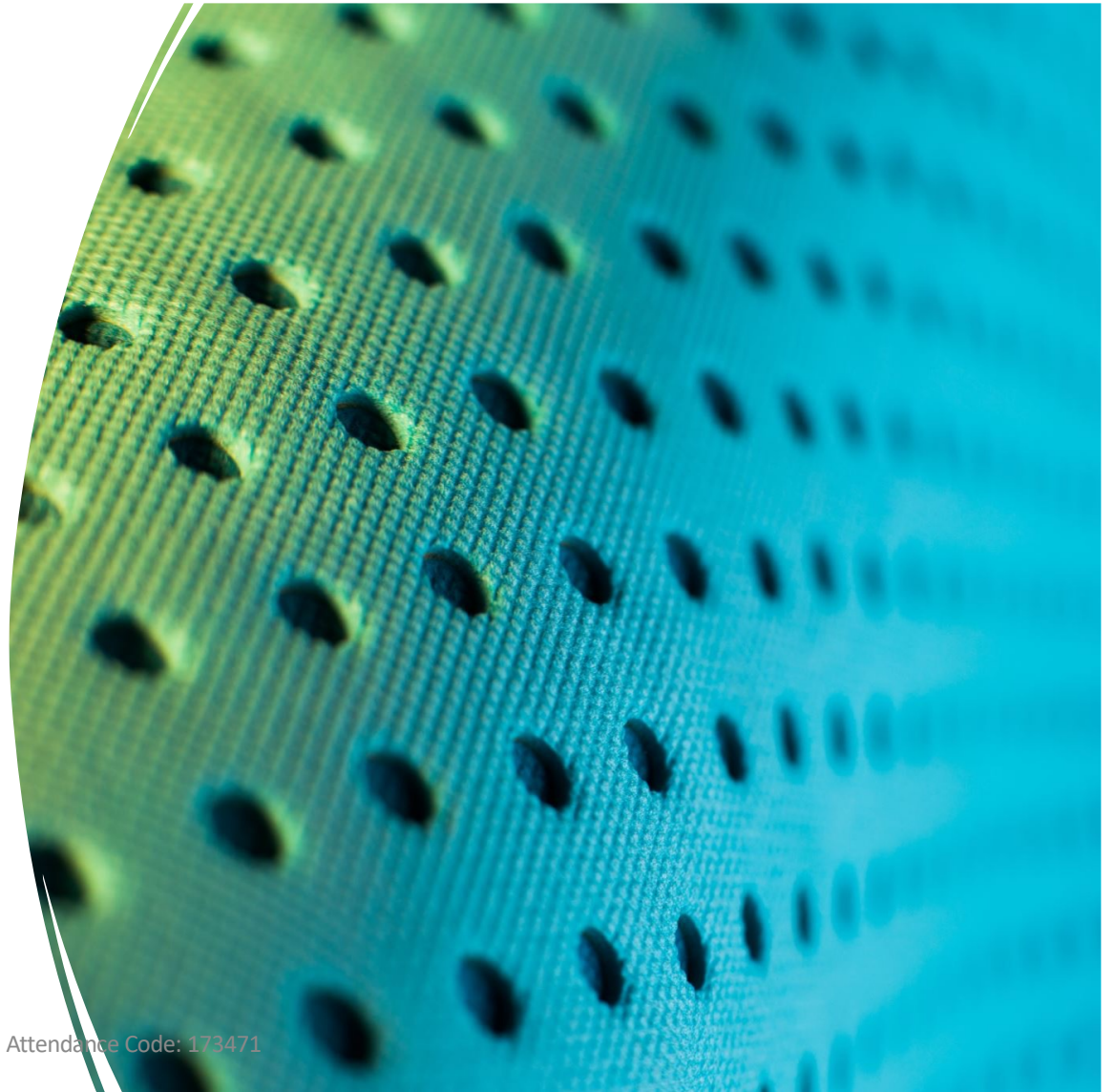
$$\nabla_x f(x) = 0 \equiv \begin{cases} \frac{\partial}{\partial x_1} f(x) = 0 \\ \dots \\ \frac{\partial}{\partial x_n} f(x) = 0 \end{cases}$$

Example

- $y = 5x_1^5 + 4x_2 + x_3^2 + 2$
- so what are the critical points?
- the gradient is $(25x_1^4, 4, 2x_3)$
- We let $25x_1^4 = 0$ and $2x_3 = 0$, so all instances whose x_1 and x_3 are 0. but $4 \neq 0$. So there is no critical point.

Method of Gradient Descent

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Method of Gradient Descent

- The gradient points directly uphill, and the negative gradient points directly downhill
- Thus, we can decrease f by moving in the direction of the negative gradient
 - This is known as the method of **steepest descent or gradient descent**
- Steepest descent proposes a new point

$$x' = x - \epsilon \nabla_x f(x)$$

- where ϵ is the **learning rate**, a positive scalar. Set to a small constant.

Choosing ϵ : Line Search

- We can choose ϵ in several different ways
- Popular approach: set ϵ to a small constant
- Another approach is called *line search*:

- Evaluate

$$f(x - \epsilon \nabla_x f(x))$$

for several values of ϵ and choose the one that results in smallest objective function value

Example: Gradient Descent on Linear Regression

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Example: Gradient Descent on Linear Regression

Linear regression: $\hat{L}(f_w) = \frac{1}{m} \sum_{i=1}^m (w^T x^{(i)} - y^{(i)})^2 = \frac{1}{m} \|Xw - y\|_2^2$

The gradient is:

$$\begin{aligned} & \nabla_w \hat{L}(f_w) \\ = & \nabla_w \frac{1}{m} \|Xw - y\|_2^2 \\ = & \nabla_w [(Xw - y)^T (Xw - y)] \\ = & \nabla_w [w^T X^T Xw - 2w^T X^T y + y^T y] \\ = & 2X^T Xw - 2X^T y \end{aligned}$$

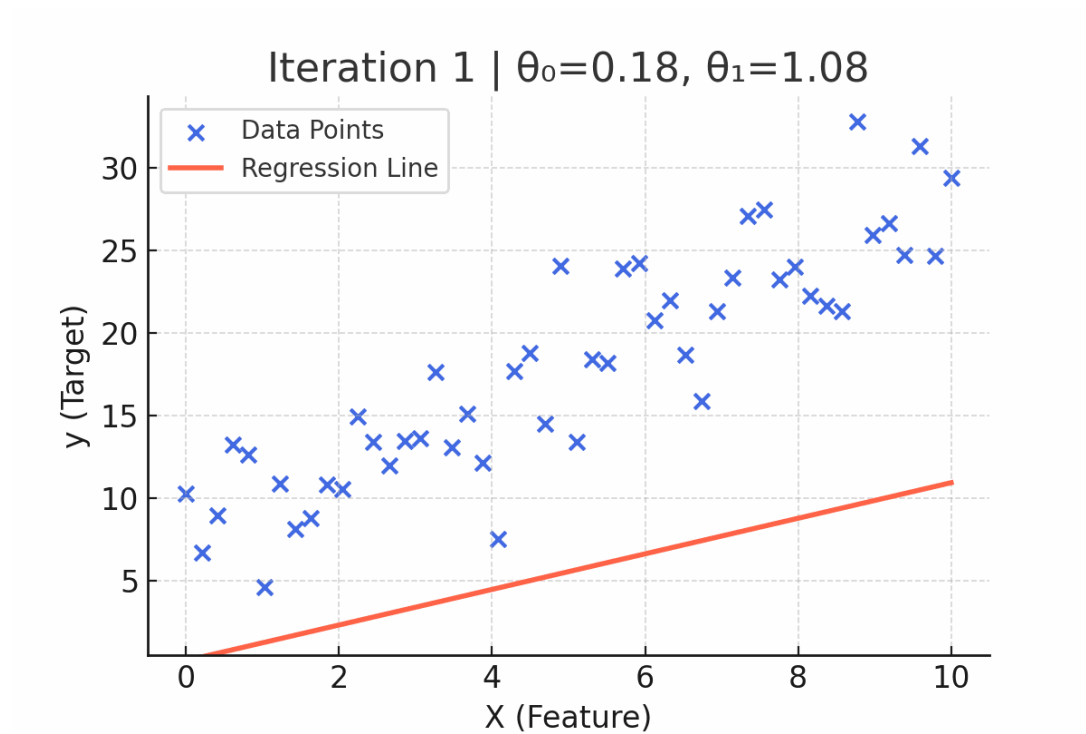
An example of designing
learning algorithms

Example: Gradient Descent on Linear Regression

- Linear regression: $\hat{L}(f_w) = \frac{1}{m} \sum_{i=1}^m (w^T x^{(i)} - y^{(i)})^2 = \frac{1}{m} \|Xw - y\|_2^2$
- The gradient is $\nabla_w \hat{L}(f_w) = 2X^T Xw - 2X^T y$
- Gradient Descent algorithm is
 - Set step size ϵ , tolerance δ to small, positive numbers.
 - *While* $\|X^T Xw - X^T y\|_2 > \delta$ *do* ——— Why? If gradient is already very small, we can terminate

$$x \leftarrow x - \epsilon(X^T Xw - X^T y)$$

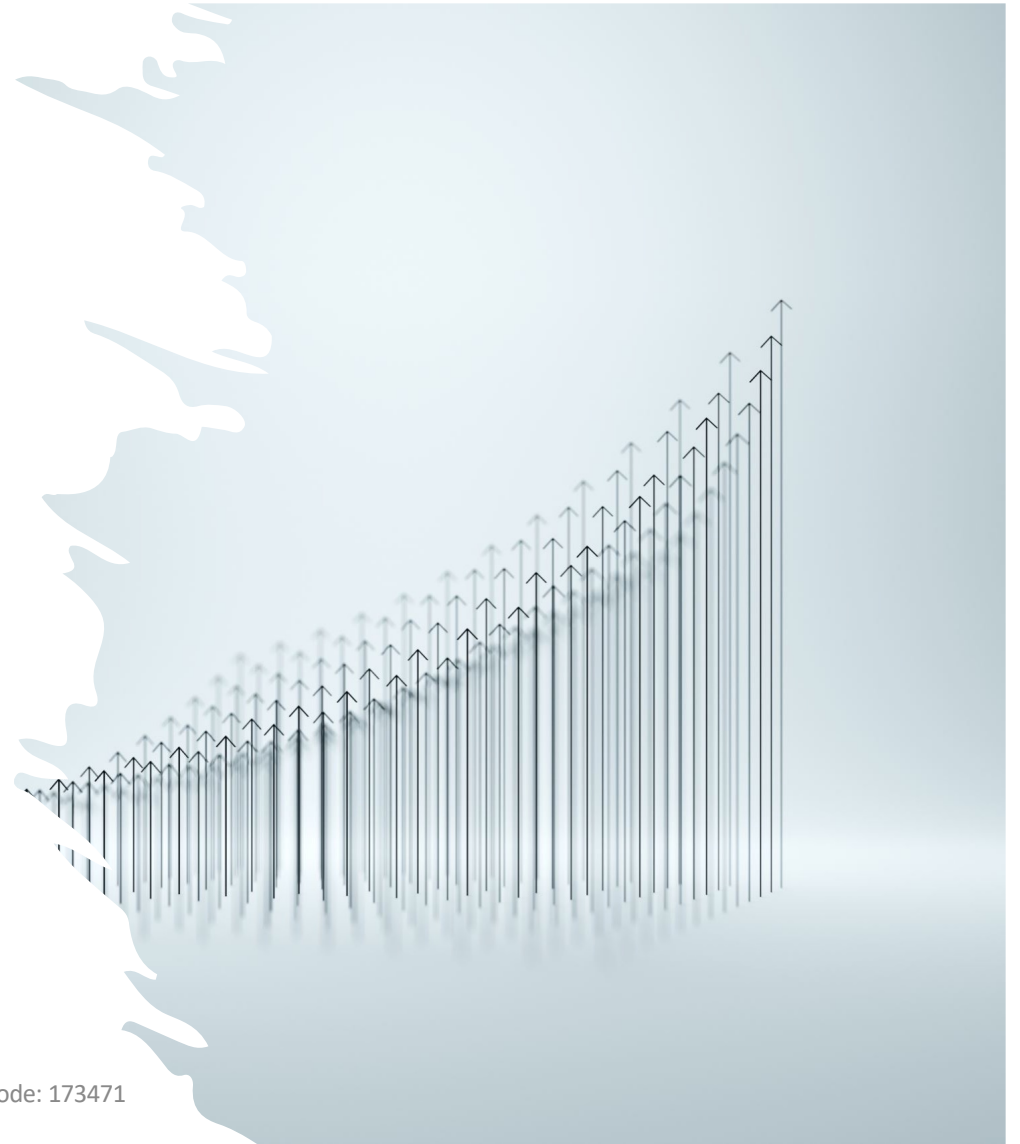
Visuals



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Linear Regression: **Analytical solution**

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Convergence of Steepest Descent

- Steepest descent converges when every element of the gradient is zero
 - In practice, very close to zero
- We may be able to avoid iterative algorithm and jump to the critical point by solving the following equation for x

$$\nabla_x f(x) = 0$$

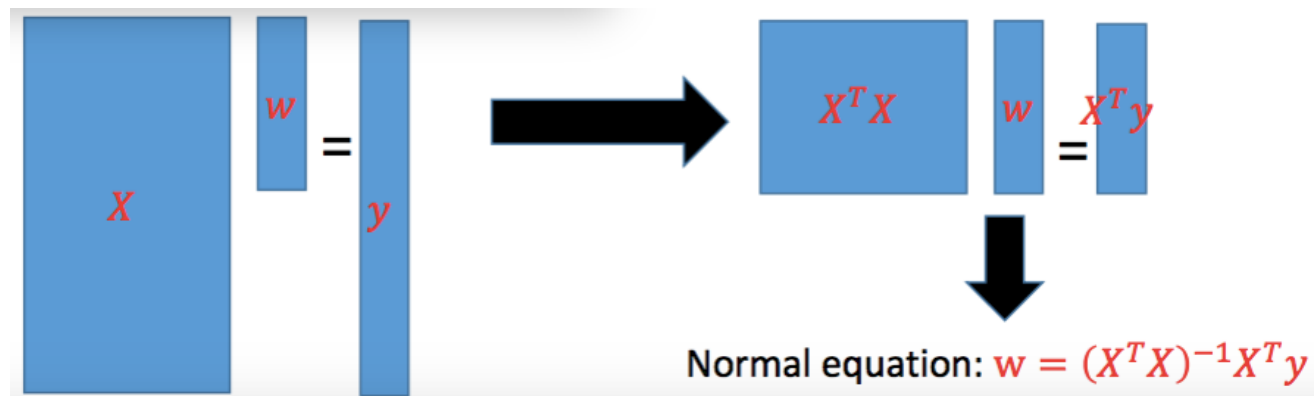
Linear Regression: Analytical solution

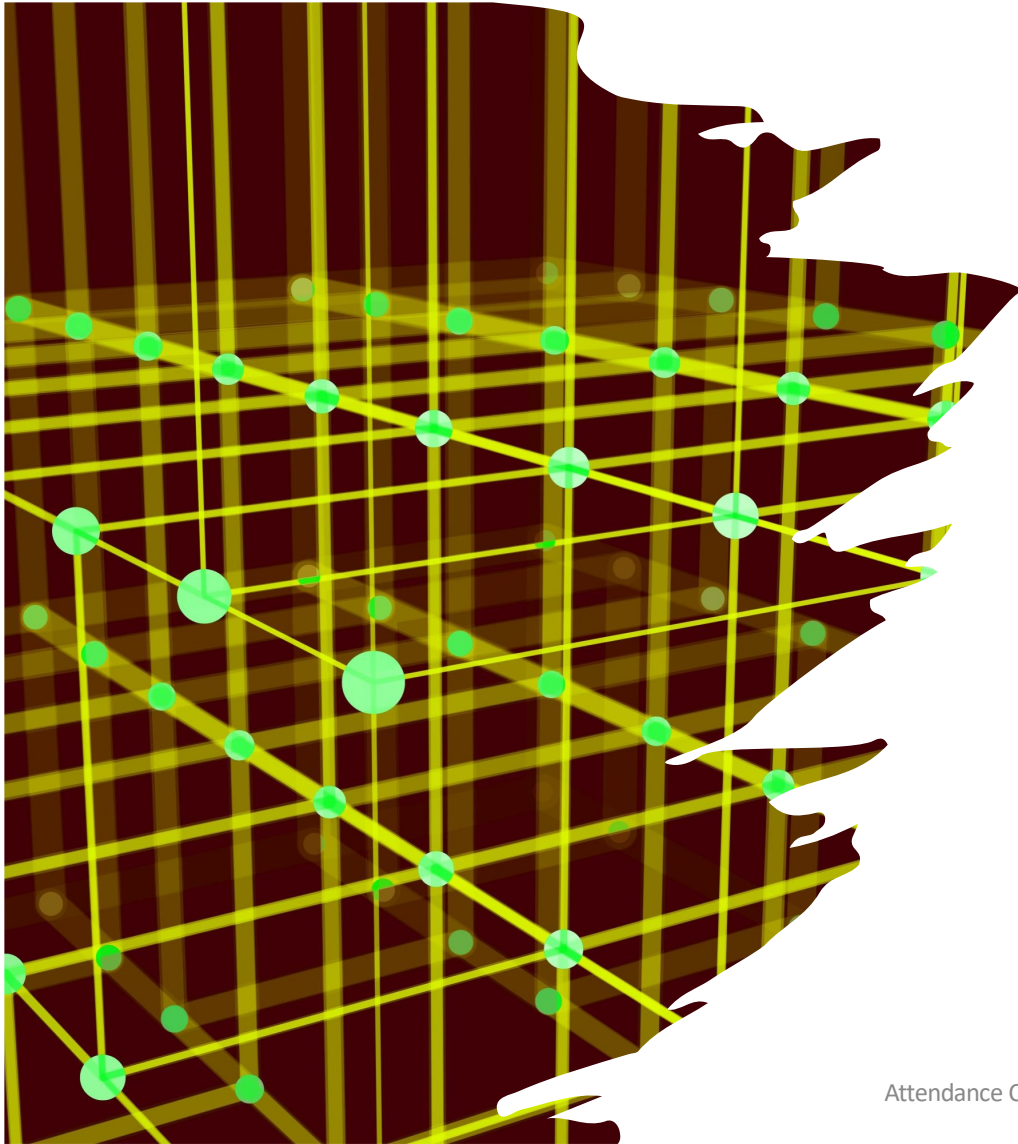
- Linear regression: $\hat{L}(f_w) = \frac{1}{m} \sum_{i=1}^m (w^T x^{(i)} - y^{(i)})^2 = \frac{1}{m} \|Xw - y\|_2^2$
- The gradient is $\nabla_w \hat{L}(f_w) = 2X^T Xw - 2X^T y$
- Let $\nabla_w \hat{L}(f_w) = 2X^T Xw - 2X^T y = 0$
- Then, we have $w = (X^T X)^{-1} X^T y$

Not all machine learning problems
have analytical solutions.

Linear Regression: Analytical solution

- Algebraic view of the minimizer
- If X is invertible, just solve $Xw = y$ and get $w = X^{-1}y$
- But typically, X is a tall matrix





Generalization to discrete spaces

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Generalization to discrete spaces

- Gradient descent is limited to continuous spaces
- Concept of repeatedly making the best small move can be generalized to discrete spaces
- Ascending an objective function of discrete parameters is called *hill climbing*

Exercises

Check the exercises of the lecture notes
for the answer to the first two questions

- Given a function $f(x) = e^x/(1+e^x)$, how many critical points?
- Given a function $f(x_1, x_2) = 9x_1^2 + 3x_2 + 4$, how many critical points?
- Please write a program to do the following: given any differentiable function (such as the above two), an ϵ , and a starting x and a target x' , determine whether it is possible to reach x' from x . If possible, how many steps? You can adjust ϵ to see the change of the answer.